

ECE 3210 quiz01

Name: _____ SOLUTIONS _____

1. (5 points) Consider the system below. Determine if this system is linear. Circle one, but also show enough work to convince me that you know what is going on.

$$y(t) = |x(t)|$$

Solution:

We can test for linearity by using some input $x(t) = ax_1(t) + bx_2(t)$ and see if the output is $y(t) = ay_1(t) + by_2(t)$. We see that

$$y(t) = |ax(t) + by(t)|$$

which cannot be manipulated to match

$$ay_1(t) + by_2(t) = a|x_1(t)| + b|x_2(t)|$$

Therefore, the system is **nonlinear**.

2. (5 points) Consider the system below. Determine if this system is time-invariant. Circle one, but also show enough work to convince me that you know what is going on.

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

Solution: Shift the input: Let $\tilde{y}(t) = \mathcal{H}\{x(t-T)\} = \int_{-\infty}^t x(\tau-T) d\tau$. We will need to do some change of variables such that $u = \tau - T$ and $du = d\tau$. We will need to adjust the limits of integration as well. When $\tau = -\infty$, $u = -\infty - T = -\infty$. When $\tau = t$, $u = t - T$. Therefore, we have

$$\tilde{y}(t) = \int_{-\infty}^{t-T} x(u) du$$

Shift the output: Now we will shift the output by T and see if we get the same result. We have

$$y(t-T) = \int_{-\infty}^{t-T} x(\tau) d\tau$$

which is the same as $\tilde{y}(t)$. Therefore, the system is **time-invariant**.